

DIABETIC RETINOPATHY SCREENING REPORT

AI-Assisted Fundus Image Analysis — with Bootstrap 95% CI

Image: idrid_IDRiD_0944.jpg
Date: 2026-03-28 12:28:36
Model: EfficientNet-B3 (Binary + Multiclass Cascade)
Bootstrap: 2000 resamples, 95% confidence level

SCREENING RESULT (with 95% Bootstrap CI)

Binary Detection:

- Ensemble (5-fold+TTA): DR Detected
P(DR) = 0.9857 95% CI [0.9834, 0.9881]
- 2-Phase (single+TTA): DR Detected
P(DR) = 1.0000 95% CI [0.9999, 1.0000]

Severity Grading (Direct Multiclass Model):

- Predicted Grade: 3 — Severe NPDR
- Confidence: 82.2% 95% CI [0.7763, 0.8684]
- Severity: High

Per-Grade Probabilities (95% CI):

G0 (No DR)	): 0.0073	[0.0040, 0.0116]
G1 (Mild NPDR)	): 0.0085	[0.0033, 0.0152]
G2 (Moderate NPDR)	): 0.1053	[0.0822, 0.1349]
G3 (Severe NPDR)	): 0.8224	[0.7763, 0.8684] ◀
G4 (Proliferative DR)	): 0.0566	[0.0336, 0.0809]

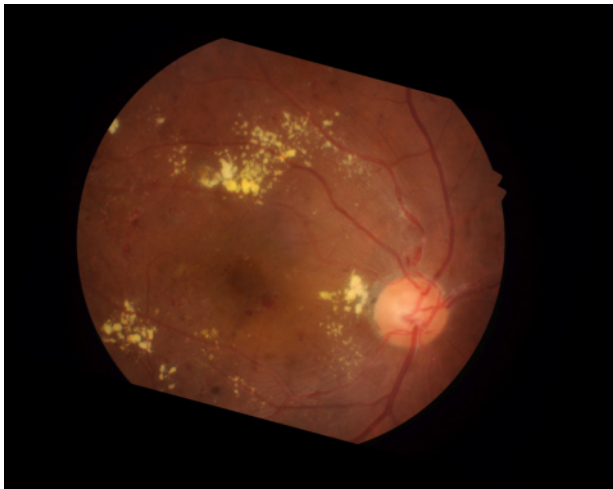
Clinical Recommendation:

- Refer to ophthalmologist within 1 month

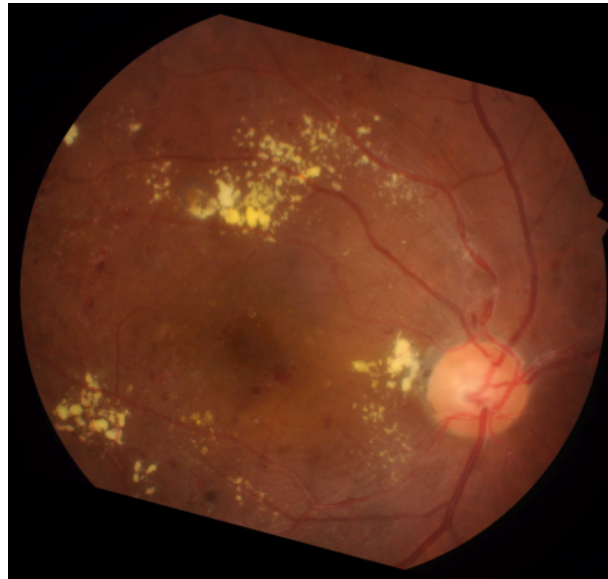
DISCLAIMER: This is an AI screening aid, not a diagnostic tool. All results must be confirmed by a qualified ophthalmologist.

Preprocessing Pipeline — idrid_IDRiD_0944.jpg

Step 1: Original



Step 2: Circle Crop



Step 3: Ben Graham



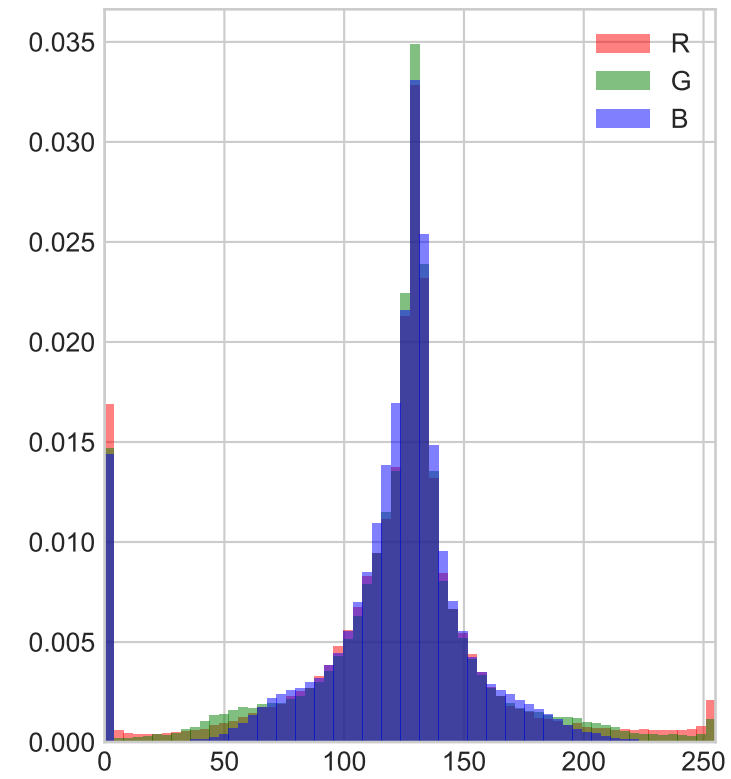
Step 4: CLAHE



Step 5: Final (384×384)

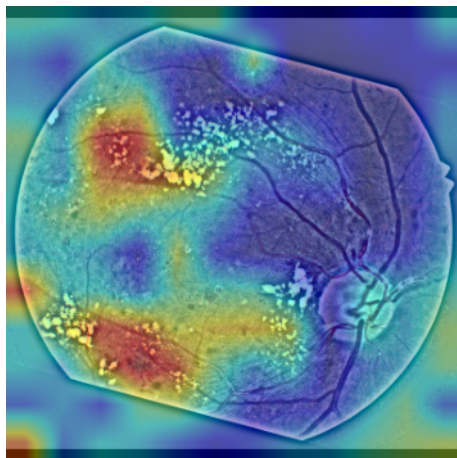


Preprocessed Histograms

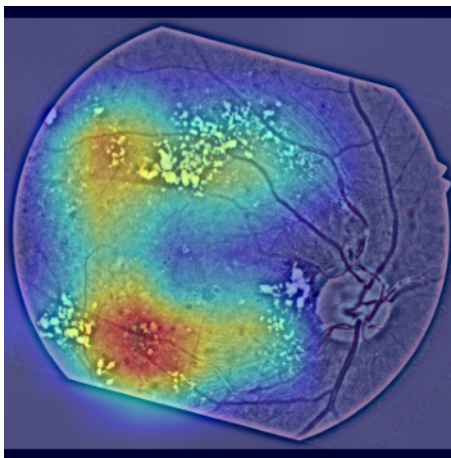


Explainability & Predictions with 95% CI — idrid_IDRiD_0944.jpg

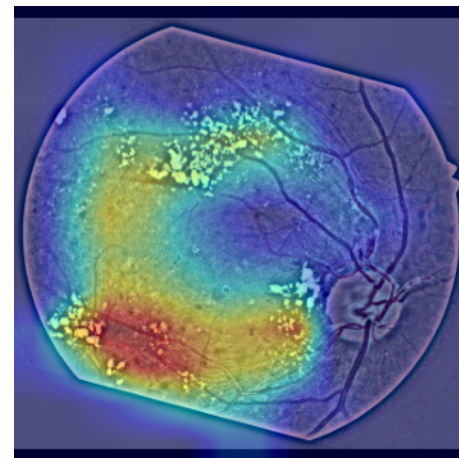
Ensemble GradCAM
P(DR)=0.9857 CI[0.9834,0.9881]



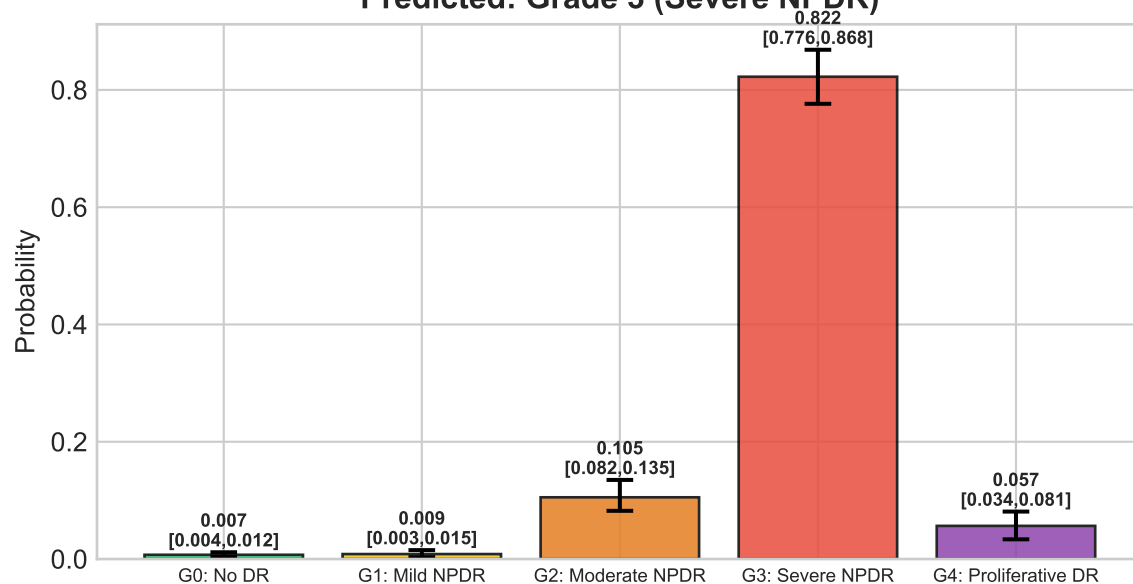
2-Phase GradCAM
P(DR)=1.0000 CI[0.9999,1.0000]



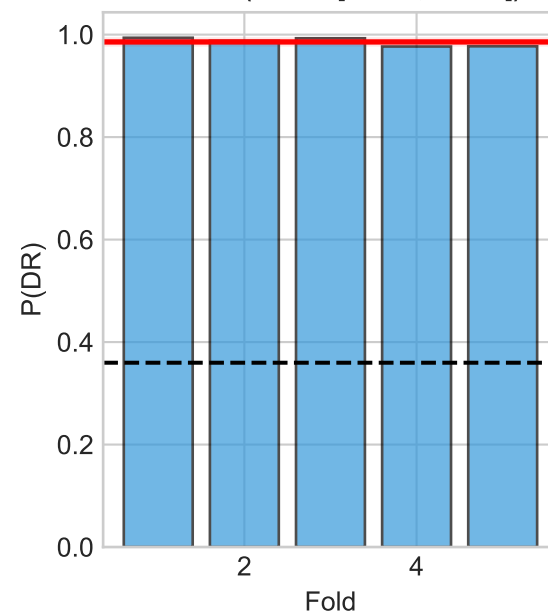
Multiclass GradCAM (G3)
Conf=82.24% CI[0.776,0.868]



5-Grade Probabilities with 95% CI (n_boot=2000)
Predicted: Grade 3 (Severe NPDR)



Per-Fold (95% CI [0.9792, 0.9923])



Model Validation Summary

MODEL VALIDATION METRICS

Dataset: APTOS 2019 (531 test images)

Bootstrap: 2000 resamples, 95% CI

Metric	Ensemble+TTA	2-Phase+TTA
AUC-ROC	0.9992	0.9992
Accuracy	0.9906	0.9887
Sensitivity	0.9962	0.9962
Specificity	0.9852	0.9815
F1-Score	0.9905	0.9886
Cohen Kappa	0.9812	0.9774

Multiclass Model (Direct 5-Class):

Best Validation QWK: 0.859

Architecture: EfficientNet-B3 + Dropout(0.4) + Linear(1536, 5)

Training: CrossEntropyLoss + label_smoothing=0.1

THIS IMAGE – PREDICTION SUMMARY

Ensemble P(DR): 0.985713 95% CI [0.983352, 0.988131]

2-Phase P(DR): 0.999952 95% CI [0.999917, 0.999977]

Multiclass Grade: 3 (Severe NPDR)

Confidence: 0.8224 95% CI [0.7763, 0.8684]

Per-Grade Probabilities (95% CI):

G0 No DR : 0.0073 [0.0040, 0.0116]

G1 Mild NPDR : 0.0085 [0.0033, 0.0152]

G2 Moderate NPDR : 0.1053 [0.0822, 0.1349]

G3 Severe NPDR : 0.8224 [0.7763, 0.8684]

G4 Proliferative DR : 0.0566 [0.0336, 0.0809]